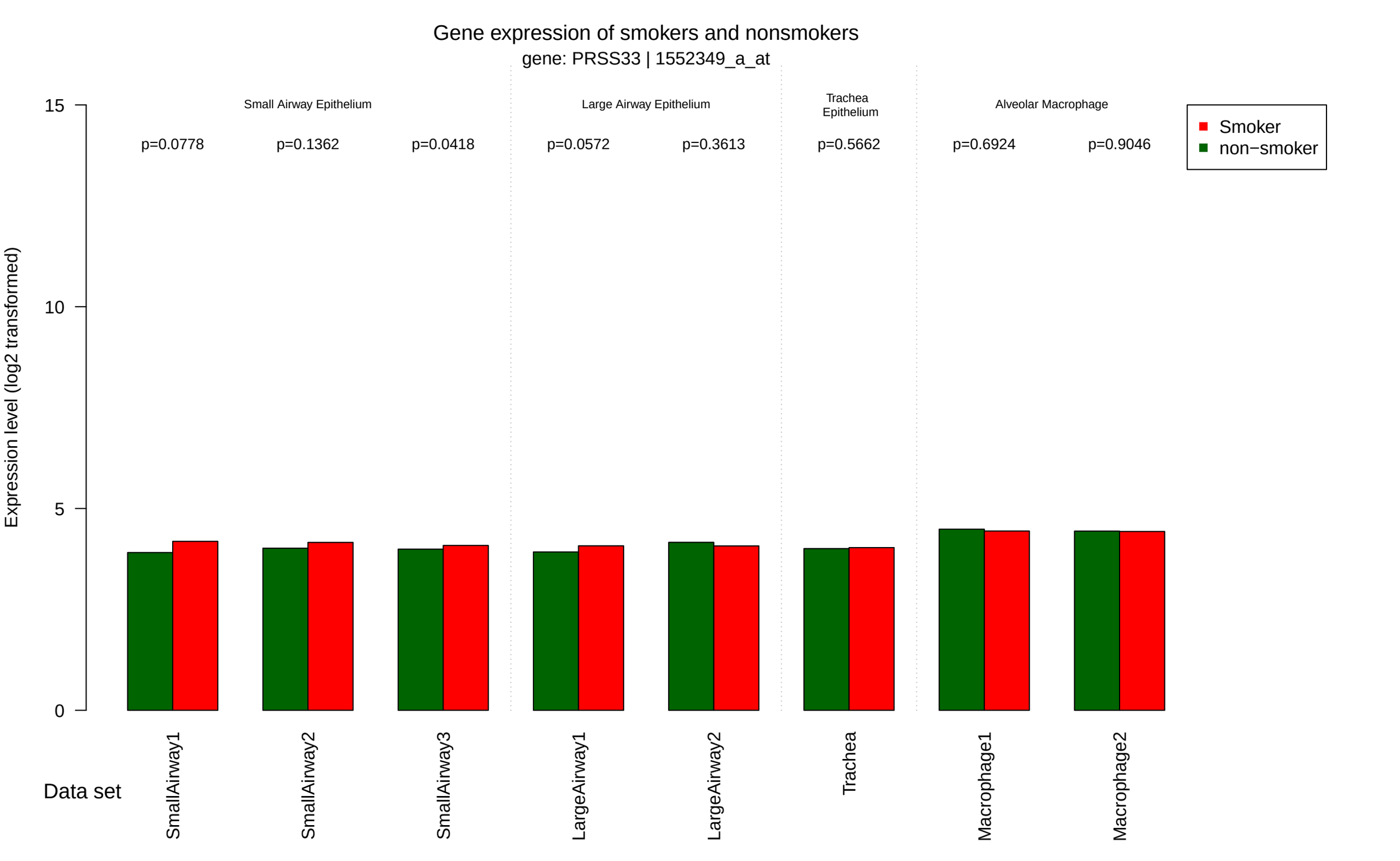
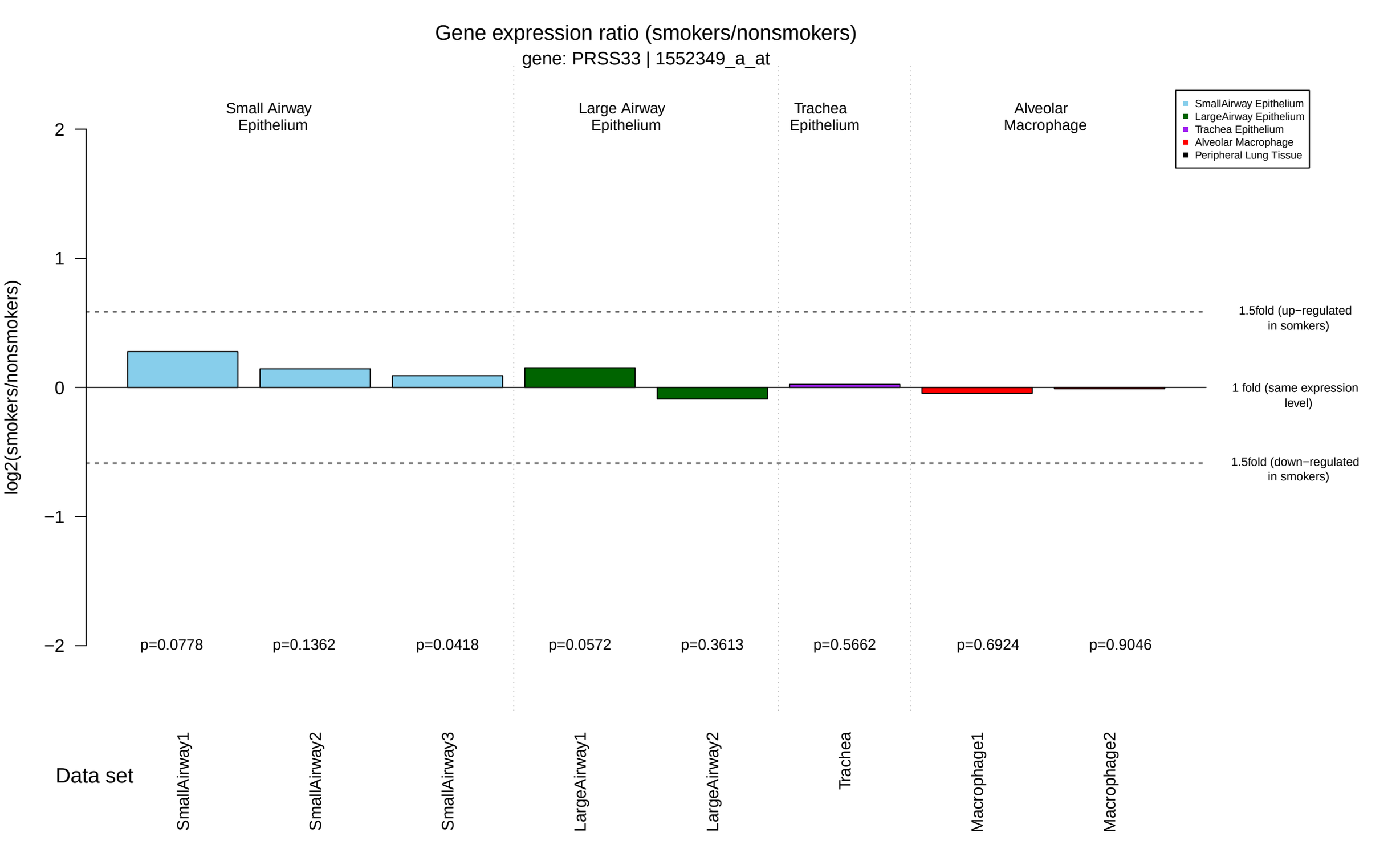
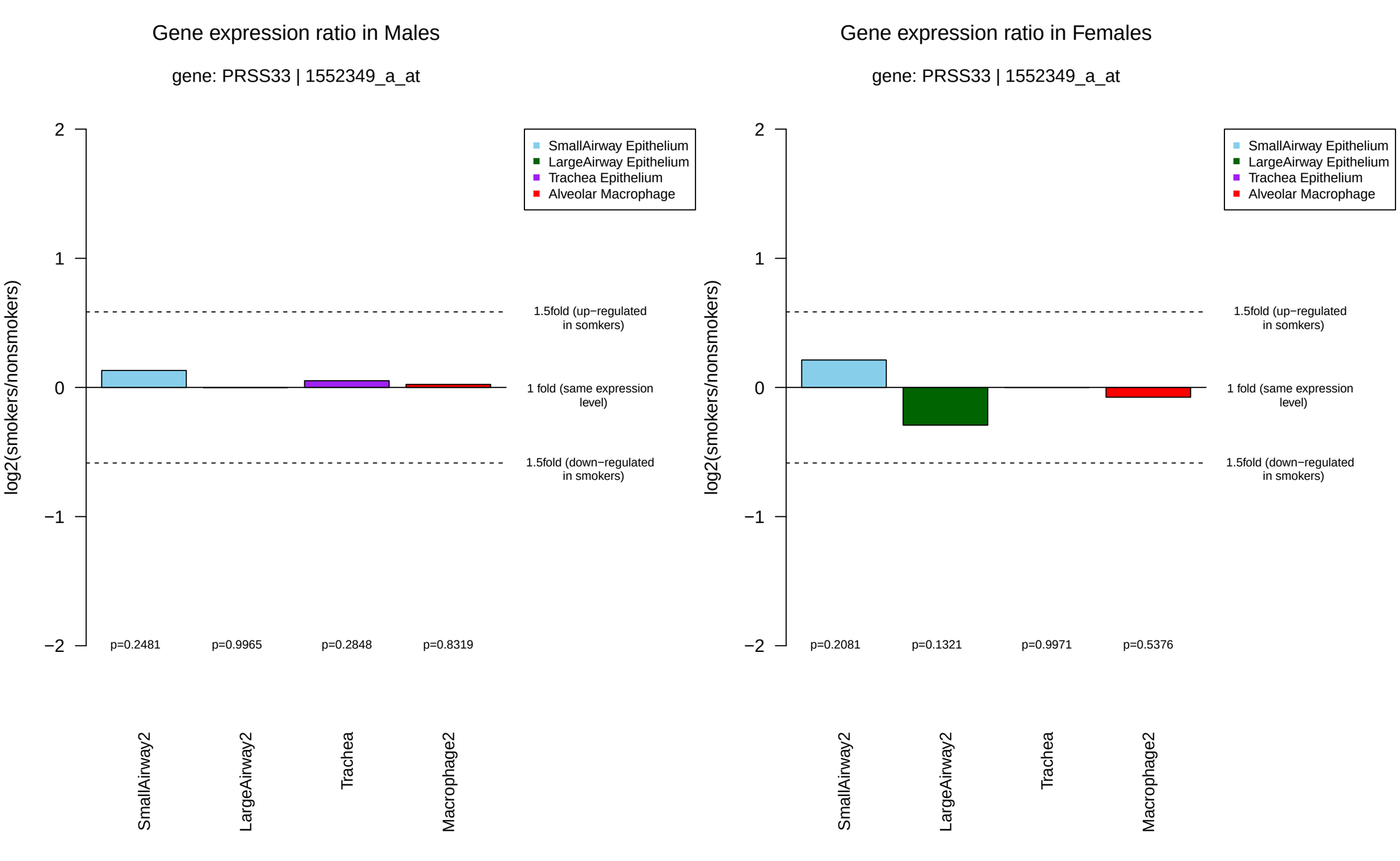
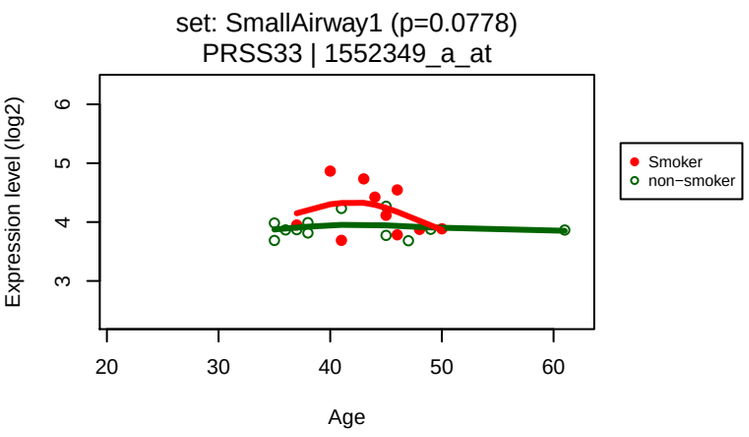
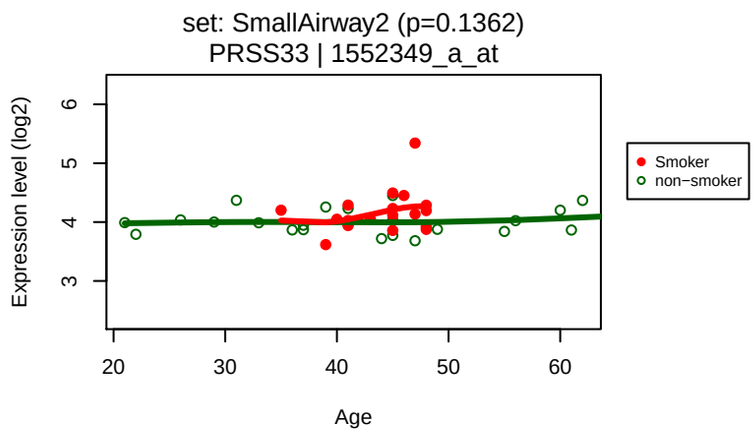
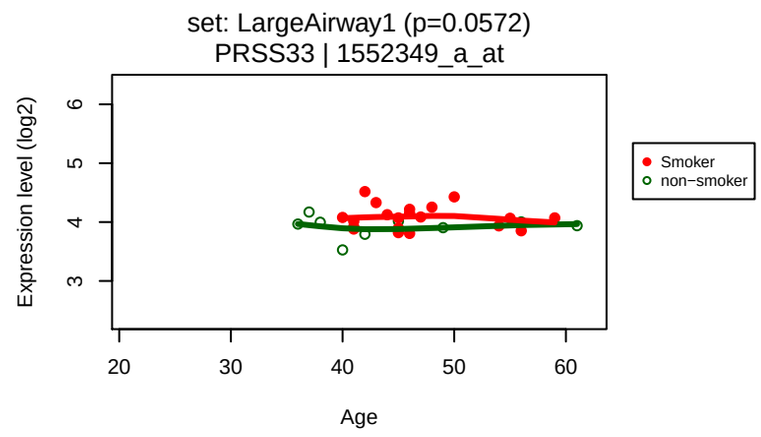

Gene expression of smokers and nonsmokers
gene: PRSS33 | 1552349_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0778
p=0.1362
p=0.0418
p=0.0572
p=0.3613
p=0.5662
p=0.6924
p=0.9046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

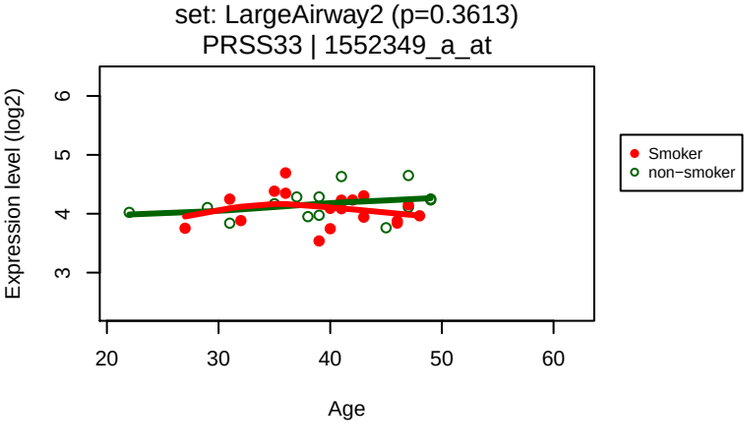

Gene expression ratio (smokers/nonsmokers)
gene: PRSS33 | 1552349_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0778
p=0.1362
p=0.0418
p=0.0572
p=0.3613
p=0.5662
p=0.6924
p=0.9046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

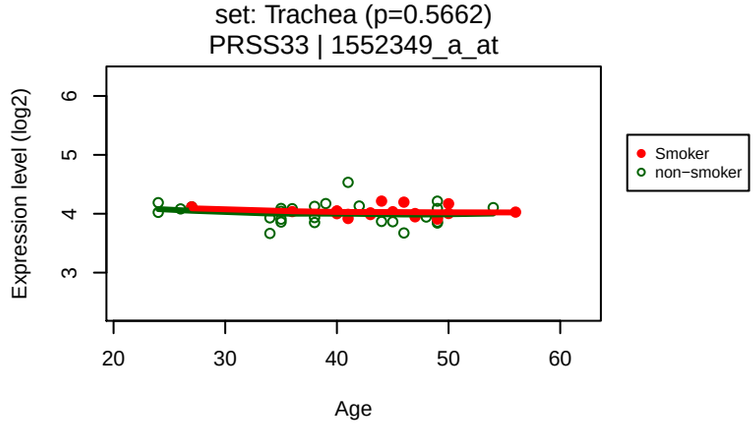

Gene expression ratio in Males
gene: PRSS33 | 1552349_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2481
p=0.9965
p=0.2848
p=0.8319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS33 | 1552349_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2081
p=0.1321
p=0.9971
p=0.5376
SmallAirway2
LargeAirway2
Trachea
Macrophage2

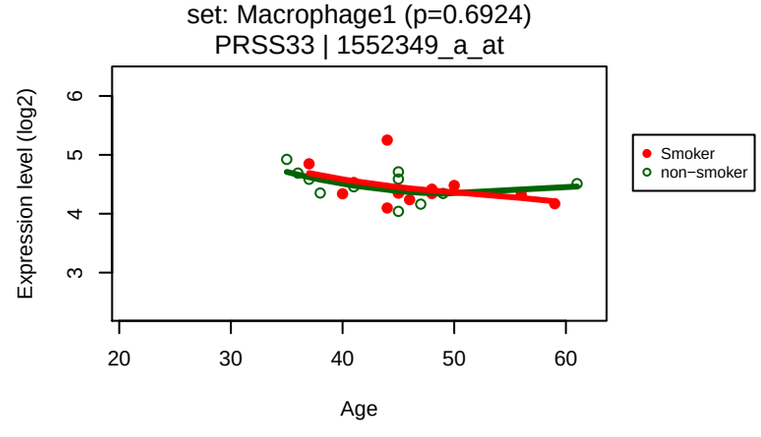

set: SmallAirway1 (p=0.0778)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker

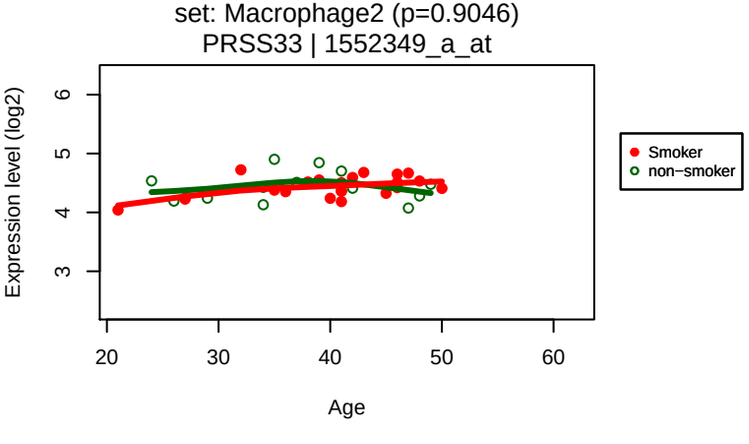

set: SmallAirway2 (p=0.1362)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker

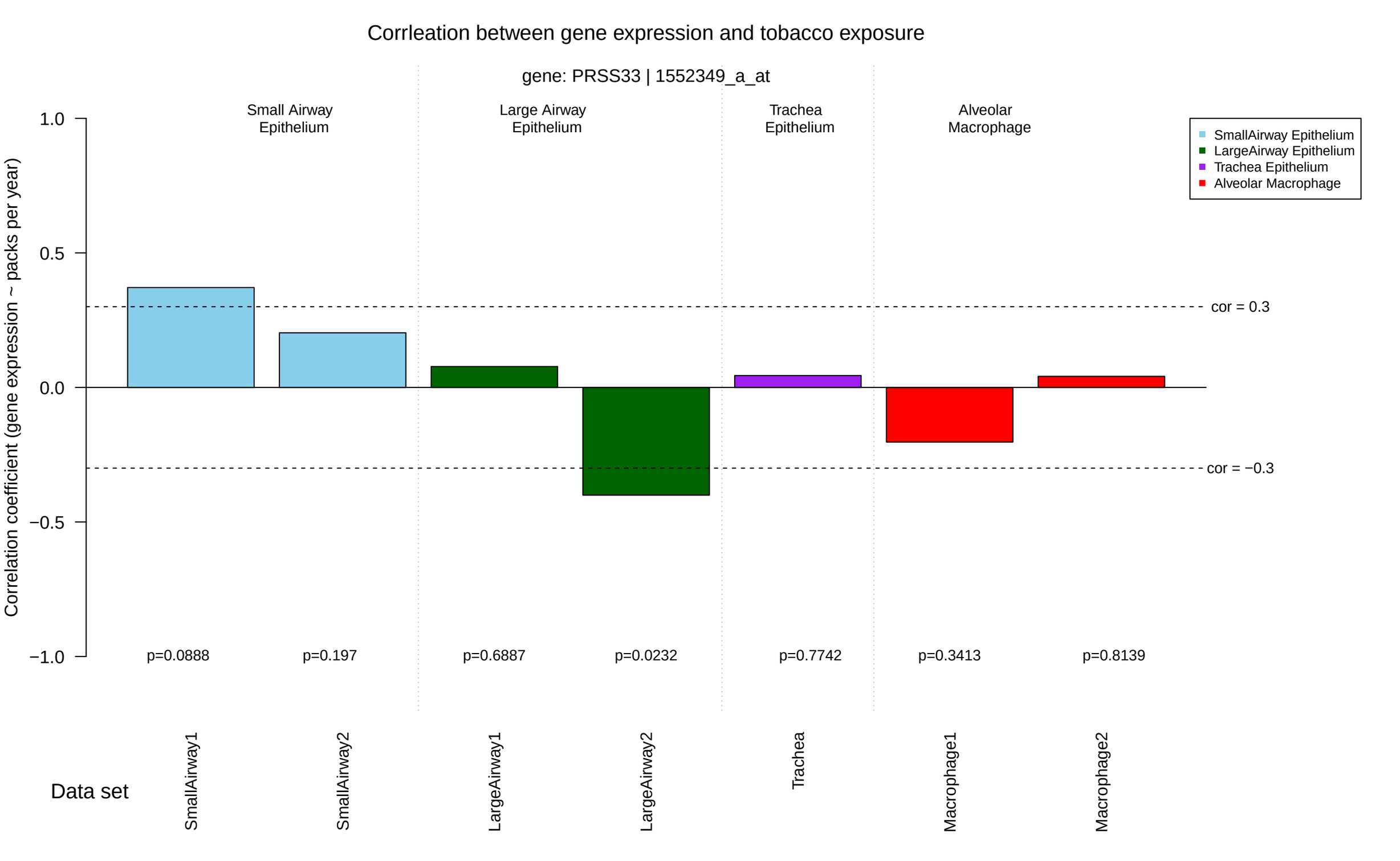

set: LargeAirway1 (p=0.0572)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3613)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5662)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6924)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9046)
PRSS33 | 1552349_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS33 | 1552349_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0888
p=0.197
p=0.6887
p=0.0232
p=0.7742
p=0.3413
p=0.8139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2